The Circles Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: MEASURE, ARC, ANGLE and LINES (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** (MEASURE)
- ☐ **A2** (MEASURE)
- ☐ **A3** (ARC)
- ☐ **A4** (ARC)
- ☐ **A5** (ANGLE)
- ☐ **A6** (LINES)

Entrées

10 pts each · choose 4

- ☐ **E1** (MEASURE)
- ☐ **E2** (ARC)
- ☐ **E3** (ANGLE)
- ☐ **E4** (ANGLE)
- ☐ **E5** (LINES)
- ☐ **E6** (LINES)

Chef's Specials

20 pts each · choose 2

- ☐ **S1** (MEASURE)
- ☐ **S2** (ARC)
- ☐ **S3** (ANGLE)
- ☐ **S4** (LINES)

My skill check (circle each skill you have covered): MEASURE · ARC · ANGLE · LINES

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 MEASURE ☐ My pick · 5 pts

A circle has radius 7 cm. Find its circumference and its area. Leave answers in terms of π.

A2 MEASURE ☐ My pick · 5 pts

A circle has circumference 20π in. Find its radius and its area, in terms of π.

A3 ARC ☐ My pick · 5 pts

Find the length of a 90° arc of a circle with radius 12 m. Leave your answer in terms of π.

A4 ARC ☐ My pick · 5 pts

Find the area of a sector with central angle 60° in a circle of radius 6 cm, in terms of π.

A5 ANGLE ☐ My pick · 5 pts

Central angle ∠AOB measures 84°, and inscribed angle ∠ACB intercepts the same arc AB. Find m∠ACB.

A6 LINES ☐ My pick · 5 pts

Point P is outside a circle with center O. Segment PT is tangent to the circle at T, OT = 5 and PT = 12. Find OP.

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 MEASURE ☐ My pick · 10 pts

A circle has center (−2, 3) and passes through the point (1, 7). Write the equation of the circle in standard form and find its area in terms of π.

E2 ARC ☐ My pick · 10 pts

A sector of a circle has area 15π cm² and central angle 150°. Find the radius of the circle and the length of the sector's arc, in terms of π.

E3 ANGLE ☐ My pick · 10 pts

Quadrilateral ABCD is inscribed in a circle. m∠A = (3x + 10)°, m∠C = (5x − 30)°, and m∠B = (4x)°. Find x, then find m∠A, m∠C and m∠D.

E4 ANGLE ☐ My pick · 10 pts

Mia says: "Inscribed angle ∠PQR intercepts arc PR, which measures 140°, so m∠PQR = 140°." Find her mistake, give the correct m∠PQR, and find the central angle ∠POR (O is the center).

E5 LINES ☐ My pick · 10 pts

Chords AB and CD intersect at point P inside a circle. AP = 2 and PB = 16. If CP = x and PD = 2x, find x and the length of CD.

E6 LINES ☐ My pick · 10 pts

From point P outside a circle, tangent segment PT = 6. A secant from P meets the circle at A (nearer) and B (farther), with PA = 4. Dev wrote: "$6^2 = 4 + PB$, so PB = 32." Find his mistake, then find the correct PB and AB.

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

Name: _____

S1 MEASURE ☐ My pick · 20 pts

A circular garden has radius 10 m. A circular pond of radius 4 m sits at its center, and a gravel path 2 m wide surrounds the garden. **(a)** Find the planted area. **(b)** Find the path area. **(c)** Gravel costs \$3 per m²; find the cost to the nearest dollar. **(d)** Eli says the path area is $\pi(2)^2 = 4\pi$. Explain his error.

S2 ARC ☐ My pick · 20 pts

A sector has central angle 72° and arc length 4π cm. **(a)** What fraction of the circle is the sector? **(b)** Find the radius. **(c)** Find the sector’s area. **(d)** Find the sector’s perimeter (arc plus two radii), in terms of π.

S3 ANGLE ☐ My pick · 20 pts

Triangle ABC is inscribed in a circle and AB is a diameter. $m\angle A = (2x + 4)°$ and $m\angle B = (3x - 9)°$. **(a)** Explain why $m\angle C = 90°$. **(b)** Find x and both angles. **(c)** Find the measures of arc BC (not containing A) and arc AC (not containing B).

S4 LINES ☐ My pick · 20 pts

From point P outside a circle with center O and radius 12, tangent segments PA and PB touch the circle at A and B. $PA = 3x - 2$ and $PB = x + 10$. **(a)** Find x and PA. **(b)** Find OP. **(c)** Find the area of quadrilateral OAPB. **(d)** Explain why $\angle OAP$ is a right angle.